

DECODELABS DATA SCIENCE INTERNSHIP

Project 2 — Fraud Detection Pipeline

Imbalanced Classification with SMOTE, Logistic Regression & Random Forest

PROGRAM Batch 2026	ROLE Data Science Intern	AUTHOR Muhammad Usaid
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Internship Project Documentation

1 OBJECTIVE

Build a leak-free classification pipeline to flag potentially fraudulent orders, defined in the supplied project documentation as Returned or Cancelled orders, using the processed dataset from Project 1. The project compares a linear baseline with an ensemble model under class imbalance.

AT A GLANCE

Best model: Logistic Regression with SMOTE, selected for stronger recall on the fraud class. The supplied documentation reports recall of 0.776 for Logistic Regression versus 0.143 for Random Forest. SMOTE is applied strictly within training folds to avoid data leakage.

2 METHODOLOGY

2.1 TARGET DEFINITION

The binary IsFraud target flags OrderStatus equal to Returned or Cancelled as 1, while all other orders are labeled 0. The supplied documentation frames this as a fraud-classification exercise based on the available proxy definition.

2.2 FEATURE ENGINEERING

Feature	Definition
AvgItemValue	TotalPrice / ItemsInCart
ItemsPerOrder	ItemsInCart / (Quantity + 1)
IsHighValue	Orders above the 95th percentile of TotalPrice
PricePerUnit	TotalPrice / Quantity
HasDiscount	Coupon usage flag
WeekendFraud	IsWeekend × IsFraud, used for exploratory analysis

2.3 TRAIN / TEST SPLIT

An 80/20 stratified train/test split is performed before oversampling. This keeps the held-out test set representative of the underlying class distribution and prevents synthetic samples from influencing evaluation.

2.4 HANDLING CLASS IMBALANCE WITH SMOTE

SMOTE is used to synthetically balance the fraud class within training data only. Both models are wrapped in an imblearn Pipeline so SMOTE is re-fit independently within each cross-validation fold.

For Logistic Regression the documented sequence is StandardScaler → SMOTE → classifier. For Random Forest the sequence is SMOTE → classifier because tree-based splits do not require feature scaling.

2.5 MODELS & TUNING

Model	Tuning
Logistic Regression	GridSearchCV over C, solver, and SMOTE k_neighbors; optimized for recall
Random Forest	GridSearchCV over n_estimators, max_depth, class_weight, and SMOTE k_neighbors; optimized for recall

Recall is used as the tuning metric rather than accuracy because missing a fraudulent order (false negative) is typically more costly than generating a false alarm.

2.6 EVALUATION

Accuracy is deliberately excluded. The supplied documentation evaluates Precision, Recall, F1-Score, and ROC-AUC instead.

3 EVALUATION RESULTS

Metric	Logistic Regression	Random Forest
Precision	0.281	0.179
Recall	0.776	0.143
F1-Score	0.413	0.159
ROC-AUC	0.649	0.550

Logistic Regression with SMOTE outperformed Random Forest on every reported metric, most notably recall: 0.776 versus 0.143. The supplied documentation notes that the stronger recall means more actual fraud-class cases are caught, at the cost of more false positives.

3.1 RANDOM FOREST FEATURE SIGNALS

The supplied documentation identifies IsDelivered, TotalPrice, PricePerUnit, and AvgItemValue as the top predictive signals from the Random Forest feature-importance analysis.

4 RECOMMENDATIONS FOR DEPLOYMENT

- Deploy the Logistic Regression plus SMOTE pipeline given its stronger recall.
- Monitor model performance weekly for drift and retrain monthly with new transaction data.
- Route high-value flagged transactions to a human-in-the-loop review step.
- Use Random Forest feature importances to prioritize manual fraud investigations.
- Document modeling decisions for audit and compliance purposes.

5 KEY TAKEAWAYS

- SMOTE was applied strictly inside the training pipeline and folds, avoiding leakage into the test set.
- Precision and recall trade-offs, rather than accuracy, drove model selection.
- Logistic Regression provided substantially stronger recall than Random Forest in the supplied run.
- Order delivery status and order-value-related features were reported as strong predictive signals.

6 TECHNICAL SUMMARY

Area	Implementation
Classification	Logistic Regression, Random Forest
Imbalance handling	SMOTE
Hyperparameter tuning	GridSearchCV
Model-selection metric	Recall
Evaluation	Precision, Recall, F1-Score, ROC-AUC
Pipeline	imblearn Pipeline with fold-level resampling
Data split	80/20 stratified split

7 CONCLUSION

The project demonstrates how an imbalanced classification workflow can be structured to reduce leakage risk and prioritize the business cost of missed positive cases. Based on the supplied documentation, Logistic Regression with SMOTE is the recommended model for deployment in this run because it achieved the strongest recall and the best overall reported metrics.

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Source: Supplied Project 2 documentation and DecodeLabs Internship repository materials.